Project Design Phase-II

Technology Stack (Architecture & Stack)

Project Name: Strategic Product Placement Analysis Using Tableau

Maximum Marks: 4 Marks

Technical Architecture

The architecture of the Strategic Product Placement Analysis system shows how retail sales data is processed and visualized using Tableau dashboards. Users interact with dashboards to explore product placement, pricing, promotions, and sales performance insights.

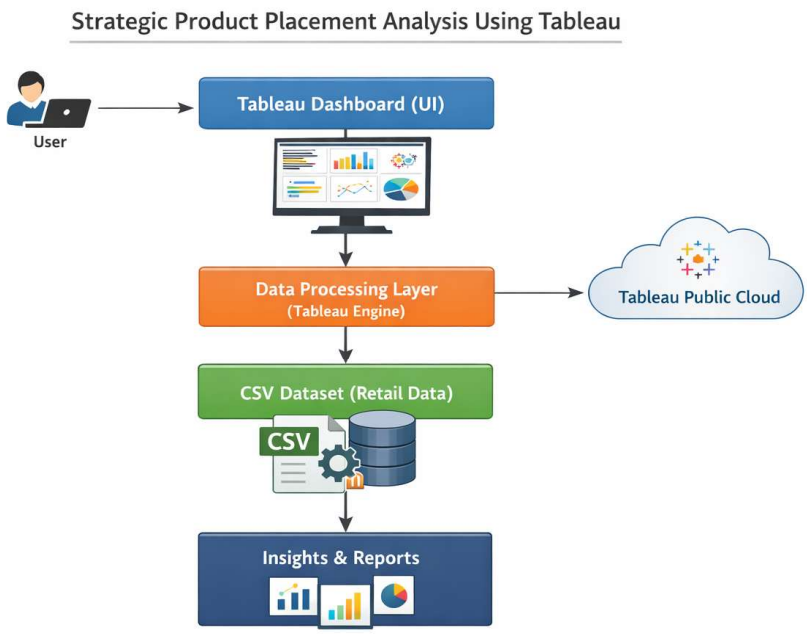

Figure: Architecture diagram for Strategic Product Placement Analysis using Tableau.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive dashboard where users analyze product sales and	Tableau Dashboard

		insights	
2	Application Logic-1	Visualization and dashboard creation	Tableau Public
3	Application Logic-2	Interactive filters and parameters	Tableau Filters & Parameters
4	Application Logic-3	Sales calculations and analytics logic	Tableau Calculated Fields
5	Database	Retail dataset storing product sales information	CSV Dataset
6	Cloud Database	Cloud environment where dashboards are published	Tableau Public Cloud
7	File Storage	Local storage of dataset files	Local File System
8	External API-1	Not used in this project	N/A
9	External API-2	Not used in this project	N/A
10	Machine Learning Model	Not used in this project	N/A
11	Infrastructure (Server / Cloud)	System used to run and publish dashboards	Local System / Tableau Public

Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Visualization platform used for data analytics	Tableau Public
2	Security Implementations	Dashboard sharing and access control	Tableau Public Permissions
3	Scalable	Handles increasing dataset size and	Tableau Data Engine

	Architecture	complex visualizations	
4	Availability	Dashboards accessible online through web platform	Tableau Public
5	Performance	Fast rendering of charts and responsive filters	Tableau Visualization Engine